

1. Database Overview

The banking database is implemented in Oracle SQL and contains 14 interconnected tables representing the bank's customers, branches, accounts, transactions, loans, employees, cards, transfers, fees, loan payments, standing orders, beneficiaries, ATMs, and audit activities. The database supports analysis of customer behavior, financial activity, loan performance, and branch performance.

2. Table Purpose

Table	Purpose
BRANCHES	Stores bank branch locations and cities
CUSTOMERS	Stores registered bank customer information
ACCOUNTS	Stores customer savings and checking accounts
TRANSACTIONS	Records deposits, withdrawals, and transfers
LOANS	Stores loans issued to customers
EMPLOYEES	Stores bank employees and management hierarchy
CARDS	Stores debit and credit cards linked to accounts
TRANSFERS	Records internal account-to-account transfers
BENEFICIARIES	Stores customers' saved payment recipients

AUDIT_LOG	Records INSERT, UPDATE, and DELETE operations
LOAN_PAYMENTS	Stores loan repayment history
STANDING_ORDERS	Stores recurring automatic payments
ATM_MACHINES	Stores ATM information and branch association

3. Primary Keys and Foreign Keys

3.1 Primary Keys

A Primary Key (PK) is a column that uniquely identifies each record in a table. Each of the 14 tables in the banking database has its own primary key.

Table	Primary Key	Purpose
BRANCHES	branch_id	Unique identifier for each branch
CUSTOMERS	customer_id	Unique identifier for each customer
ACCOUNTS	account_id	Unique identifier for each account
TRANSACTIONS	transaction_id	Unique identifier for each transaction
LOANS	loan_id	Unique identifier for each loan
EMPLOYEES	employee_id	Unique identifier for each employee

CARDS	card_id	Unique identifier for each card
TRANSFERS	transfer_id	Unique identifier for each transfer
BENEFICIARIES	beneficiary_id	Unique identifier for each beneficiary
FEES	fee_id	Unique identifier for each fee record
AUDIT_LOG	log_id	Unique identifier for each audit log entry
LOAN_PAYMENTS	payment_id	Unique identifier for each loan payment
STANDING_ORDERS	order_id	Unique identifier for each standing order
ATM_MACHINES	atm_id	Unique identifier for each ATM

3.2 Foreign Keys

A Foreign Key (FK) is a column that references the primary key of another table. Foreign keys connect tables together and allow related information to be retrieved using SQL joins.

Child Table	Foreign Key	Parent Table	Referenced Key
CUSTOMERS	branch_id	BRANCHES	branch_id
ACCOUNTS	customer_id	CUSTOMERS	customer_id
TRANSACTIONS	account_id	ACCOUNTS	account_id
LOANS	customer_id	CUSTOMERS	customer_id
EMPLOYEES	branch_id	BRANCHES	branch_id

EMPLOYEES	manager_id	EMPLOYEES	employee_id
CARDS	account_id	ACCOUNTS	account_id
TRANSFERS	from_account_id	ACCOUNTS	account_id
TRANSFERS	to_account_id	ACCOUNTS	account_id
BENEFICIARIES	customer_id	CUSTOMERS	customer_id
FEES	account_id	ACCOUNTS	account_id
LOAN_PAYMENTS	loan_id	LOANS	loan_id
STANDING_ORDERS	account_id	ACCOUNTS	account_id
STANDING_ORDERS	beneficiary_id	BENEFICIARIES	beneficiary_id
ATM_MACHINES	branch_id	BRANCHES	Branch_id

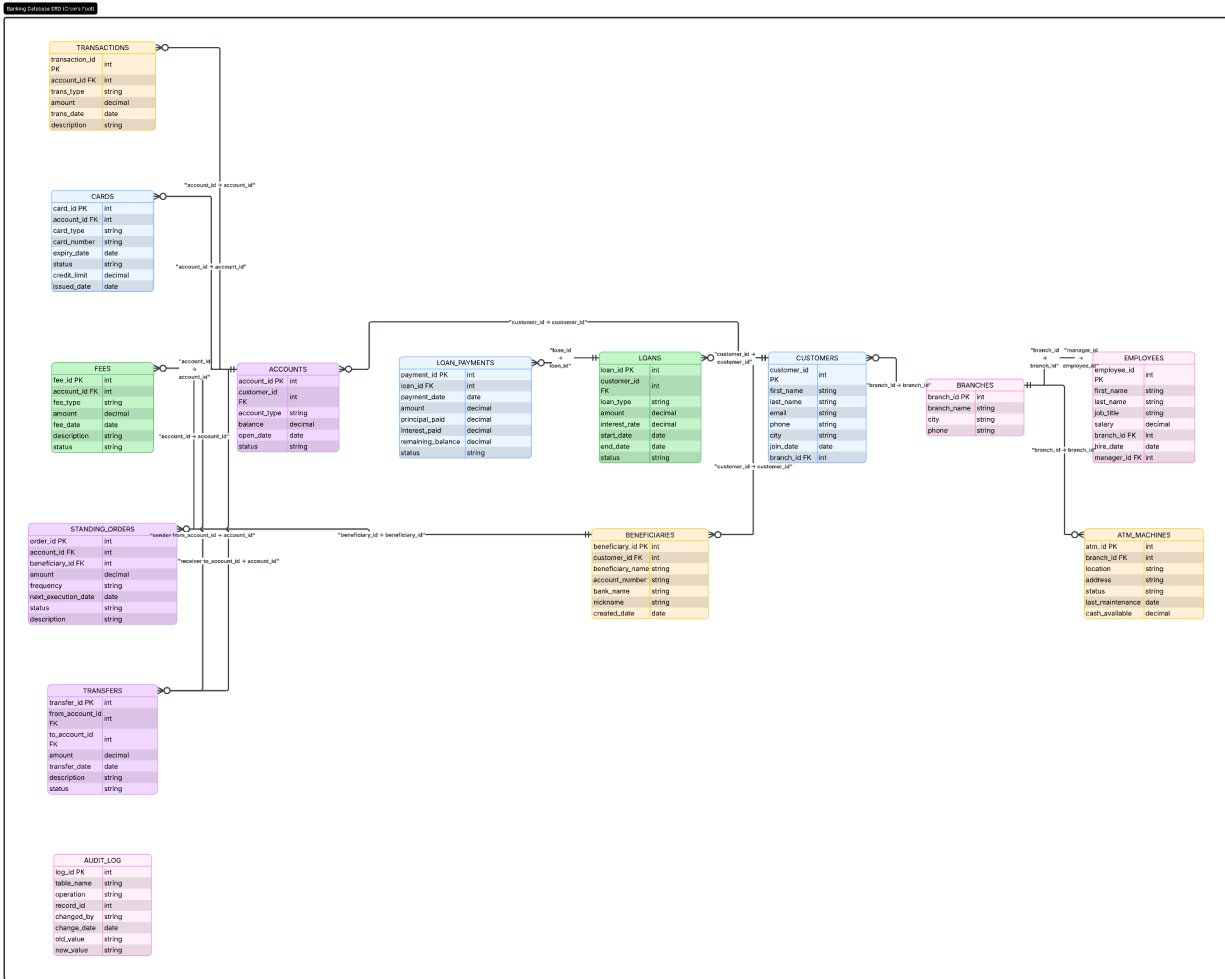
ER Diagram

The Entity Relationship Diagram (ERD) for the banking database was created using Lucidchart. The diagram illustrates the database tables, primary keys (PK), foreign keys (FK), and the relationships between the different entities using Crow's Foot notation.

The complete interactive ER diagram can be accessed through the following Lucidchart link:

Lucidchart ER Diagram:

<https://lucid.app/lucidchart/0bfa27dd-f087-4949-ad1c-44bdf24b8034/edit?viewp ort loc=-1274%2C-3947%2C3010%2C1575%2C0 0&invitationId=inv c306d006-ab d5-43a0-9ef7-2c97ab345d92>



BRANCHES is one of the main parent tables. A branch can serve multiple customers, employ multiple employees, and manage multiple ATM machines. Each customer belongs to a branch. Customers may own multiple accounts, take multiple loans, and save multiple beneficiaries. Accounts are central to financial activity because transactions, cards, fees, transfers, and standing orders are linked to accounts. Loans are connected to customers, while LOAN_PAYMENTS records the repayment history associated with each loan.